## Cell Biology & Biochemistry

**Topics Covered**
**Biochemistry**
Biomolecules and Molecular Organization
Biological Macromolecules:
- Carbohydrates
- Lipids
- Proteins
- Nucleic acids and Nucleotides
- Identification of Biomolecules (Practical)

**Cell Biology**
- Cell ultra-structure
- Biological membranes: Structure and functions
- DNA Replication
- Protein Biosynthesis
- Cell Division
- Cell signaling

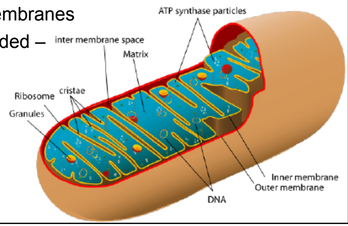
Bioenergetics
- Enzymes: General properties and mechanism of action
- Cellular respiration

# BIO 1201
# Cell Biology & Biochemistry
## Lesson 6 – Cell Ultrastructure

## Cell Biology

**Eukaryotic cells**

Other cellular organelles – Mitochondria
- Tubular organelles
- Size: 0.5 – 1.0 μm
- Found in all types of eukaryotic cells
- Bounded by two membranes
- Inner membrane folded – **cristae**




## Cell Biology

**Eukaryotic cells**

Other cellular organelles – Mitochondria
- Cristae partition the mitochondrion in to two compartments
  - matrix: lies in the inner membrane
  - intermembrane space: space between the two membranes
- On the surface of the inner membrane – there are proteins that carry out **oxidative phosphorylation**

## Cell Biology

**Eukaryotic cells**

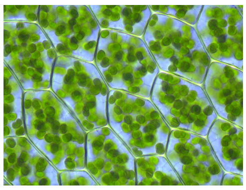
Other cellular organelles – Mitochondria

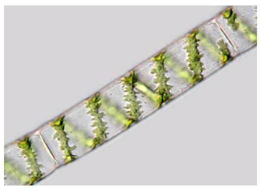
- Have their own genetic material (mitogenome) - circular DNA strands
- Mitogenome bears **genes that code for proteins used in oxydative phosphorylation**
- Proteins are synthesized within the mitochondrion
- 95% of the proteins required for mitochondrial metabolism are coded by nuclear genes and are synthesized in the cytoplasm
- Proteins synthesized in the cytoplasm are imported in to the mitochondrion
- During cell division, each mitochondrion in the cell divide in to two and double in number
- Proteins needed for mitochondrial division are coded in the nucleus

## Cell Biology

**Eukaryotic cells**

Other cellular organelles – Chloroplasts
- Found in photosynthesizing cells i.e. plants, algae etc.
- Generally lens shaped
- Size - 5–8 μm in diameter and 1–3 μm thick
- Number in a cell is variable



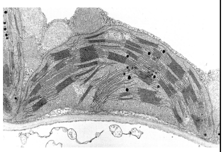


## Cell Biology

### Eukaryotic cells

Other cellular organelles – Chloroplasts

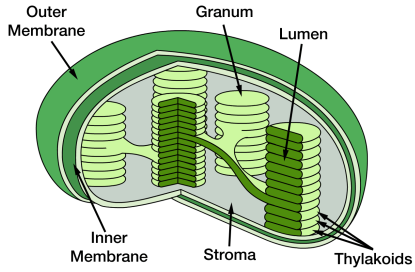
- Bound by two membranes
- Have closed compartments of stacked membranes – **grana** (may contain a >100)
- Each granum may contain several dozen disk shaped structures – **thylakoids**
- Pigments required for photosynthesis are found on the surface of the thylakoids
- Fluid matrix that surrounds the grana – **stroma**
- Contain circular DNA – chloroplast genome



## Cell Biology

### Eukaryotic cells

Other cellular organelles – Chloroplasts




## Cell Biology

### Eukaryotic cells

Other cellular organelles – Chloroplasts

- About 30 genes in the chloroplast genome code for proteins that are used in photosynthesis
- These proteins are synthesized in the chloroplast
- Bears ribosomes for protein synthesis
- 90% of the chloroplast proteins are coded by the nucleus, some are also involved in photosynthesis
- They are synthesized in the cytoplasm & are imported in to the chloroplast
- When deprived of light for a long time, internal structure changes – leucoplasts
- Leucoplasts store starch in roots
- New chloroplasts are made from proplastids or division

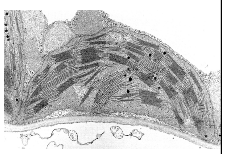
## Cell Biology

### Eukaryotic cells

Other cellular organelles – Chloroplasts

**Function**

- Conversion of $CO_2$ to carbohydrates (Photosynthesis)
- Production of energy (ATP)
- Synthesis of amino acids, fatty acids
- Conversion of $NO_2^-$ to $-NH_2$



## Cell Biology

### Eukaryotic cells

Other cellular organelles

**Origins of Mitochondria and Chloroplasts - Endosymbiosis**

Lynn Margulis

Endosymbiotic theory: mitochondria and chloroplasts descended from free living prokaryotes that were taken inside an eukaryote (Sagan, 1967)

- Symbiosis – close relationship between organisms of two species that live together

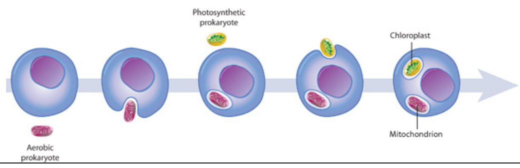
## Cell Biology

### Eukaryotic cells

Other cellular organelles

**Origins of Mitochondria and Chloroplasts – Endosymbiosis**

- One species of prokaryote was engulfed by an eukaryote and lived inside it
- Engulfed prokaryotes provided their hosts with certain advantages



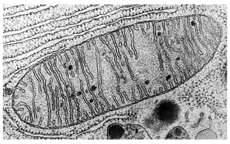
## Cell Biology

**Eukaryotic cells**

Other cellular organelles

**Origins of Mitochondria and Chloroplasts – Endosymbiosis**

What are the similarities between mitochondria, chloroplasts and prokaryotes? Membranes, size, ribosome, DNA, reproduction



## Cell Biology

**Eukaryotic cells**

Other cellular organelles – Centrosome

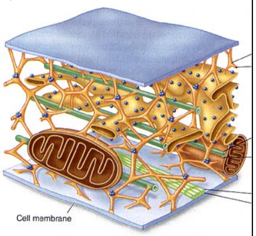
- Consists of two barrel shaped organelles occur in pairs in right (90˚) angles
- Each structure is called a **centriole**
- Made of protein **tubulin**
- Found in animal cells and protists
- Located near the nuclear membrane
- Assemble microtubules – long hollow cylinders of tubulin
- Also known as microtubule organizing centers (MTOC)
- Microtubules
  - influence cell shape
  - move chromosomes during cell division
  - provide the internal structure of flagella, cilia

## Cell Biology

**Eukaryotic cells**

Cytoskeleton

- Network of crisscrossed protein fibers that supports the shape of the cell and anchors organelles to fixed locations
- Found in all eukaryotic cells
- Dynamic system – constantly forms & dissemble
- Individual fibers – form by spontaneously assembling in to long chains

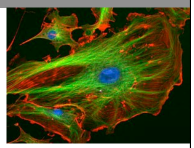


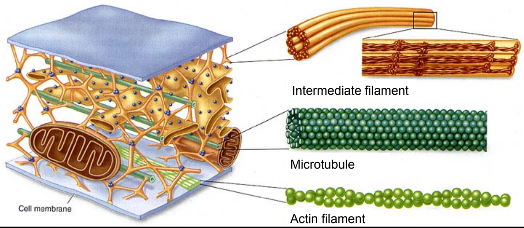

## Cell Biology

**Eukaryotic cells**

Cytoskeleton

- Consists of three types of fibers
  - Actin filaments
  - Microtubules
  - Intermediate filaments





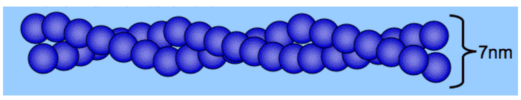

## Cell Biology

**Eukaryotic cells**

Cytoskeleton

<u>Actin filaments</u>

- Long fibers, 7 nm in diameter
- Each filament consists of two protein chains loosely tied together
- Each subunit of the chain – globular protein '**actin**'
- Responsible for cellular movements – contraction, crawling, pinching, formation of cellular extensions (cytokinesis, phagocytosis)

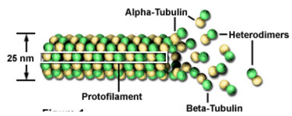



## Cell Biology

**Eukaryotic cells**

Cytoskeleton

<u>Microtubules</u>




- Hollow tubes, 25 nm in diameter
- Each tube composed of 13 protein filaments (protofilament)
- Each subunit of the filament – globular protein α and β 'tubulin'
- Form from MTOC's from the center of the cell and radiate towards the periphery (except in plants)
- Continuously form and break
- Responsible - cellular movements, moving materials within the cell, moving, spindles that move chromosomes

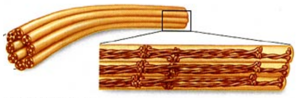
## Cell Biology

**Eukaryotic cells**

Cytoskeleton

Intermediate filaments

- Long tough fibers, 8-10 nm in diameter
- Twined together in an overlapping arrangement
- Each subunit of the chain – protein 'vimentin' or 'keratin'
- Stable and do not break down
- Provides – structural stability for cells, holds organelles in the cell

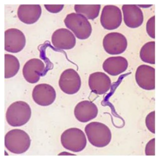


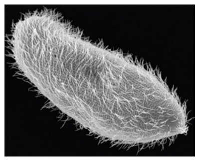
## Cell Biology

**Eukaryotic cells**

**Flagella and Cilia**

- Both outward projections of the cell's interior, containing cytoplasm enclosed by the plasma membrane
- Flagella – longer projections of the cell membrane, few in a cell
- Cilia- shorter projections of the cell membrane, many per cell





## Cell Biology

**Eukaryotic cells**

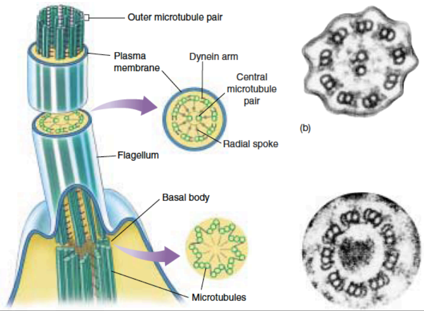
**Flagella and Cilia**

- Both eukaryotic flagella and cilia have the same interior structure
- Consists of a circle of nine microtubule pairs surrounding two central microtubules - **9 + 2 structure**
- Microtubules are derived from a **basal body,** situated just below the point where the flagellum/cilium protrudes from the surface of the cell

## Cell Biology

**Eukaryotic cells**

**Flagella and Cilia**




## Cell Biology

**Cell Wall**

- Only found in plant cells, bacteria, fungi and some protists
- Protect and support the cell
- Plant cell walls are composed of cellulose fibers
- In plants, cell walls have three strata/layers

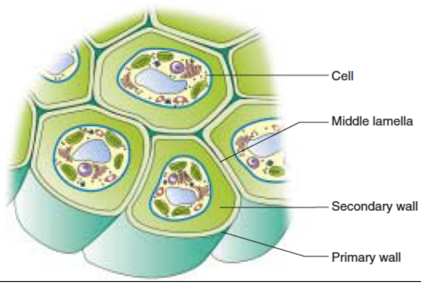
  **Primary cell wall**- thin, flexible and extensible layer formed while the cell is growing, consists of cellulose, hemicellulose fibers in a pectin matrix

  **Secondary cell wall**- thick layer formed inside the primary cell wall after the cell is fully grown, consists of hemicellulose and cellulose, cells in xylem contain **lignin**

## Cell Biology

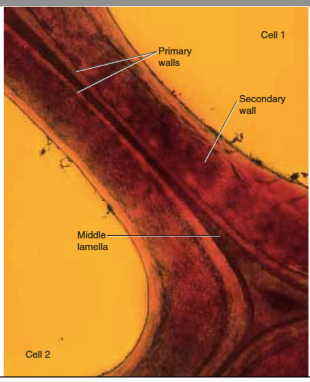
**Cell Wall**

**Middle lamella**- outermost layer, consists of **pectin**, glues the cells together



## Cell Biology

**Cell Wall**




## Cell Biology

**Summary**

- Mitochondria are double membrane bound organelles that carry out **oxidative phosphorylation** in the cell
- Chloroplasts are double membrane bound organelles that carry out photosynthesis
- According to the endosymbiotic theory mitochondria and chloroplasts descended from free living prokaryotes
- Centrosomes assemble microtubules
- Cytoskeleton is a network of crisscrossed protein fibers that supports the shape of the cell and anchors organelles
- **Cell walls** which are found in plant cells, bacteria, fungi and some protists protect and support the cell
- Flagella and Cilia are outward projections of the cell's interior, containing cytoplasm enclosed by the plasma membrane